COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions — each order's full history is auditable through ledger entries and notifications — and a role-separated workflow. Customers order; staff fulfil; drivers deliver; administrators report. Deployment is local-only: one host, all roles in one app (§4). Stock checks, confirmation transaction and role gates guard integrity. All stored data is course demo seed data; no real personal data is processed.

2. Development process

An increment is a unit of work closed by a review or a recorded test run.

Table A. Increments

Increment and date	Trigger	Outcome
Planning, 9–11 Oct 2026	Coursework kickoff	Scope, responsibilities and contracts — agreed endpoint and data shapes — recorded (increment register)
17 Oct 2026 review	Order and stock functions ready for review	Defects recorded in the increment register; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Review found a confirmation endpoint that accepted repeat submits and a missing ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Order, stock, driver and admin functions complete	Application integrated; all four role workflows runnable end to end
Driver and privacy, 2 Nov 2026	Driver assignment tests and data-minimisation checks outstanding	Driver assignment and data-minimisation checks pass (tests)
Diagram and traceability review, 10–11 Nov 2026	Review found diagrams and traceability diverging from the implementation	Diagrams and traceability aligned (diagram review record)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date CSV export in three in-app interface interactions; the 11 Nov regression exercised all four role workflows (browser checks)
CI health tests, 11–12 Nov 2026	The record of CI-run outcomes (observations) had no regression test	Two regression tests added; suite reached 38 (unit.log)

3. Requirements

Functional requirements

FR	The system shall	Verification
FR-01	create orders with product lines and totals	Test
FR-02	enforce lifecycle and stock reservation/release	Test (cancel tests)
FR-03	number orders and shipments and show owner tracking	Test
FR-04	show inventory, stock-movement ledger and low-stock alerts	Test
FR-05	record order-event notifications and allow reading	Test
FR-06	provide admin aggregates and CSV export	Test
FR-07	authenticate and enforce role/object access	Test
FR-08	assign shipped orders and record delivery	Test

- NFR-01 performance — API p95 (95th percentile) ≤2 s, first contentful paint p95 ≤3 s (measurement, §6). - NFR-02 CI health — green build, test and browser runs per CI run (observations). - NFR-03 least privilege — role gates; the delivery address is released only to the assigned driver (privacy tests). - NFR-04 interaction budget — every core operation (eleven, listed in §8) is reachable in at most three in-app interface interactions; one interaction is a fill, selection or click, and login and navigation are session setup (walkthrough). - NFR-05 process quality — green CI (observations).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 schedules	scheduling plan	register
RK-02 contract rework	Constant tests	transition tests
RK-03 test week	Traceability	scope priorities
RK-04 setup	NFR-05	CI workflow
RK-05 demo data	Seed tests	seed regression test
RK-06 coursework squeeze	Weekly re-planning	risk plan
RK-07 stack	Setup guide	setup guide
RK-08 video deadlines	Weekly re-planning	risk plan
RK-09 multi-warehouse or hosted access beyond the course demo	Accepted: out of course scope (§1)	risk plan

4. System models

The four layers — pages, routes, handlers, Prisma/SQLite persistence (ORM over the SQLite database) — separate concerns. Figures 1–5 answer, in order: what each role can do; how orders transition and stock changes; how entities relate; how the confirmation transaction proceeds; how the layers connect. Each figure links its PlantUML source.

Each item references one product; each order has one or more items; each order has at most one shipment; products accumulate ledger entries; notifications relate to at most one order.

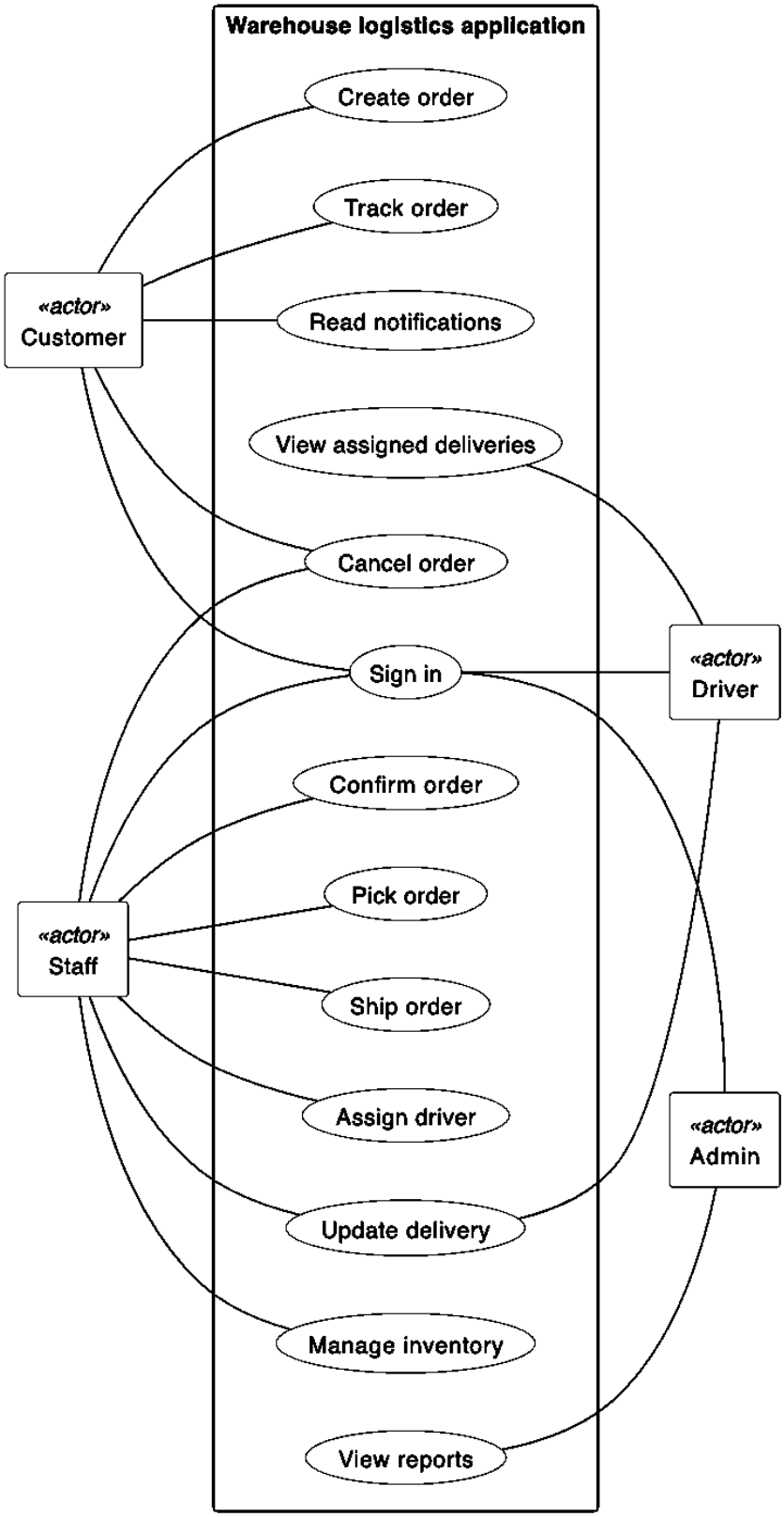


Figure 1. Role functions: thirteen use cases grouped under four role actors. Source.

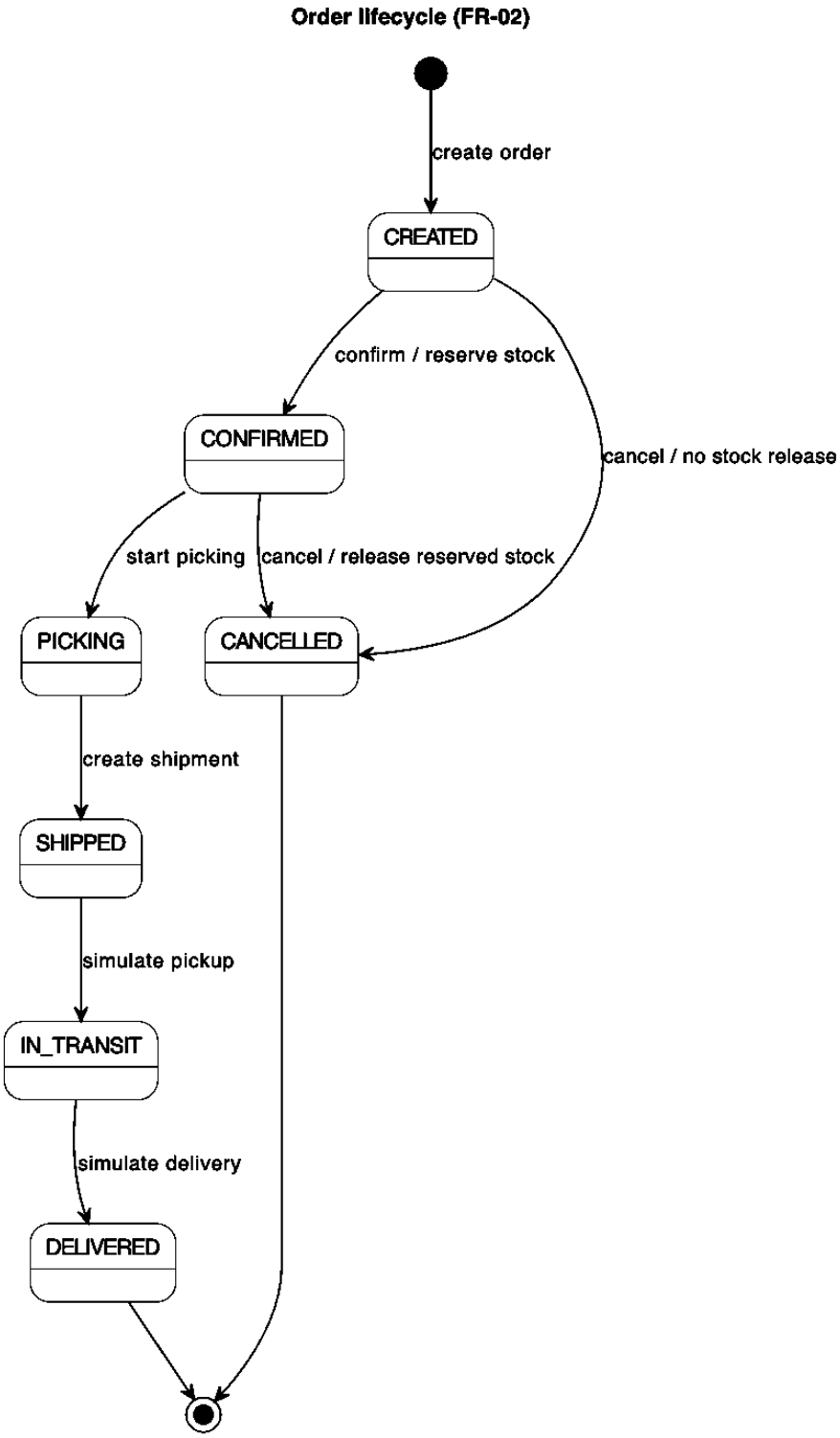


Figure 2. Order transitions and stock effects: each transition states its stock effect — reserve, release or none. Source.

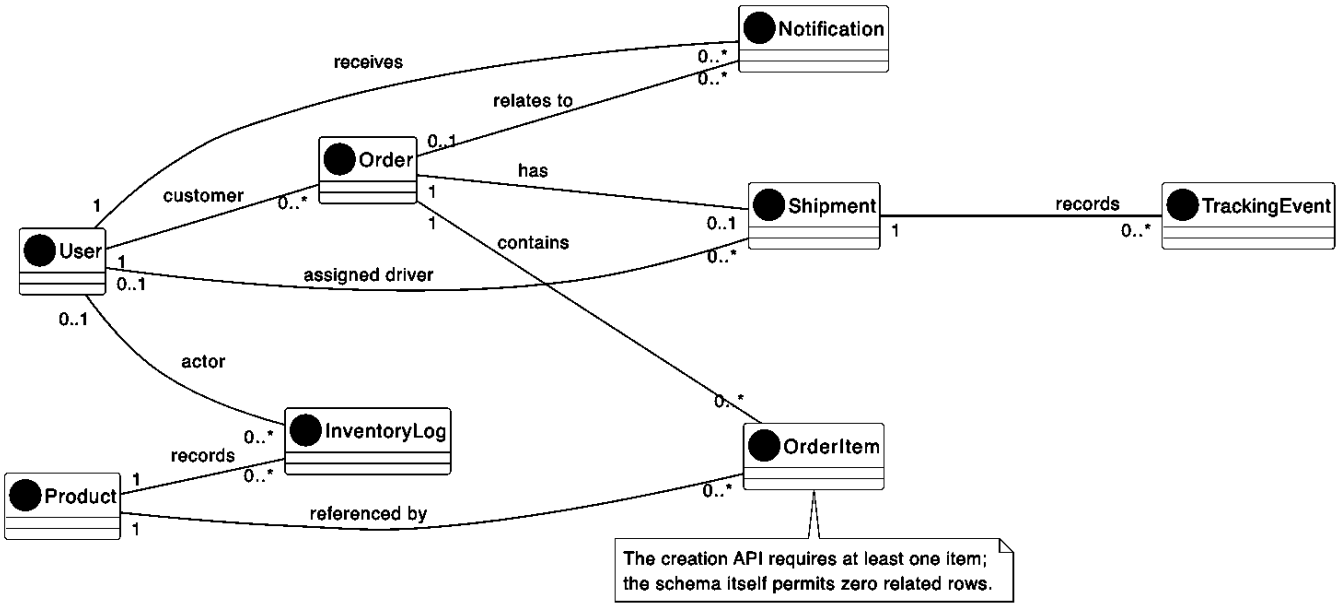


Figure 3. Data relationships: User, Order, OrderItem, Product, Shipment, TrackingEvent, InventoryLog and Notification with their multiplicities, drawn as an ER model. Source.

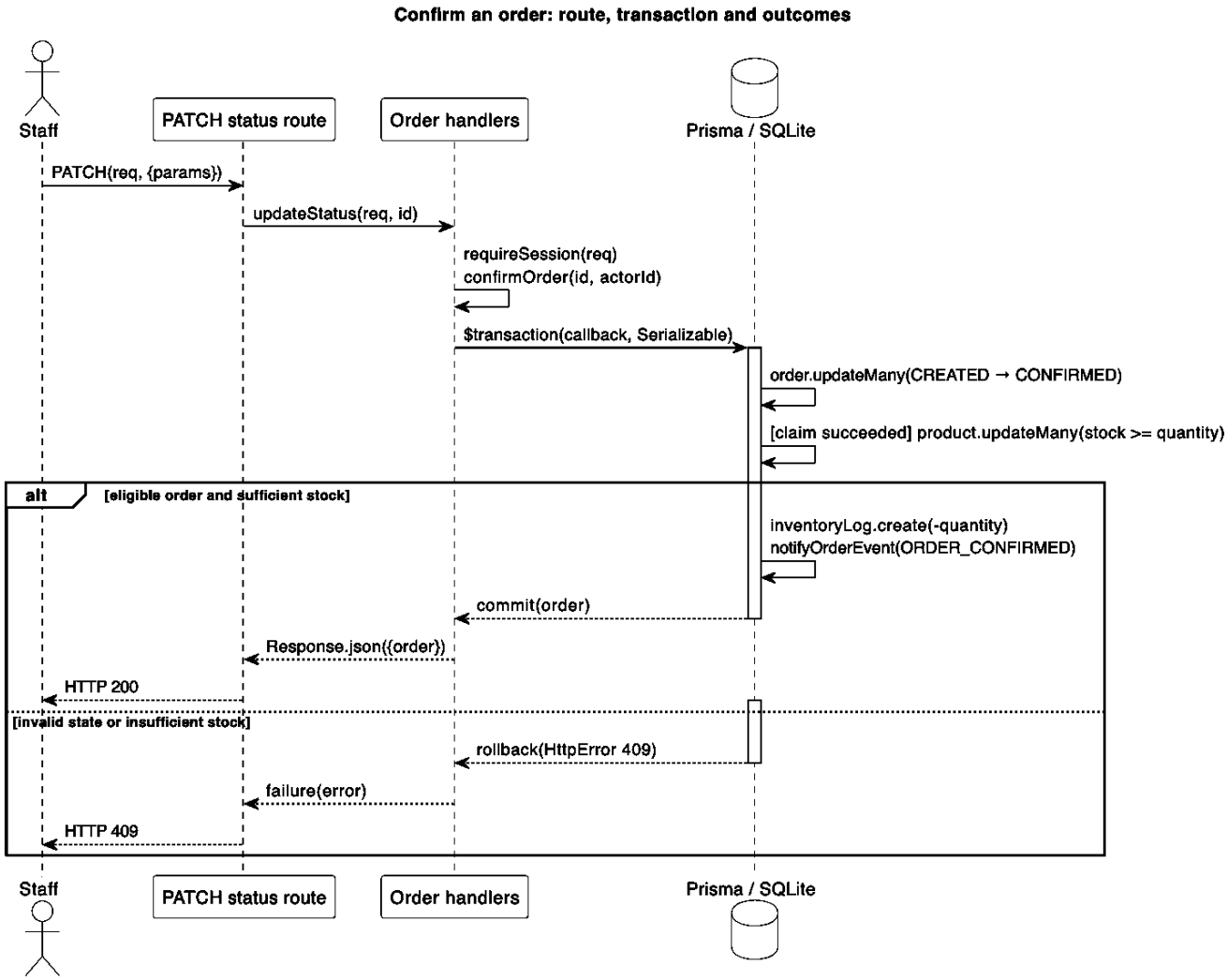


Figure 4. Confirmation transaction: session guard, conditional writes, rollback on failure or short stock. Source.

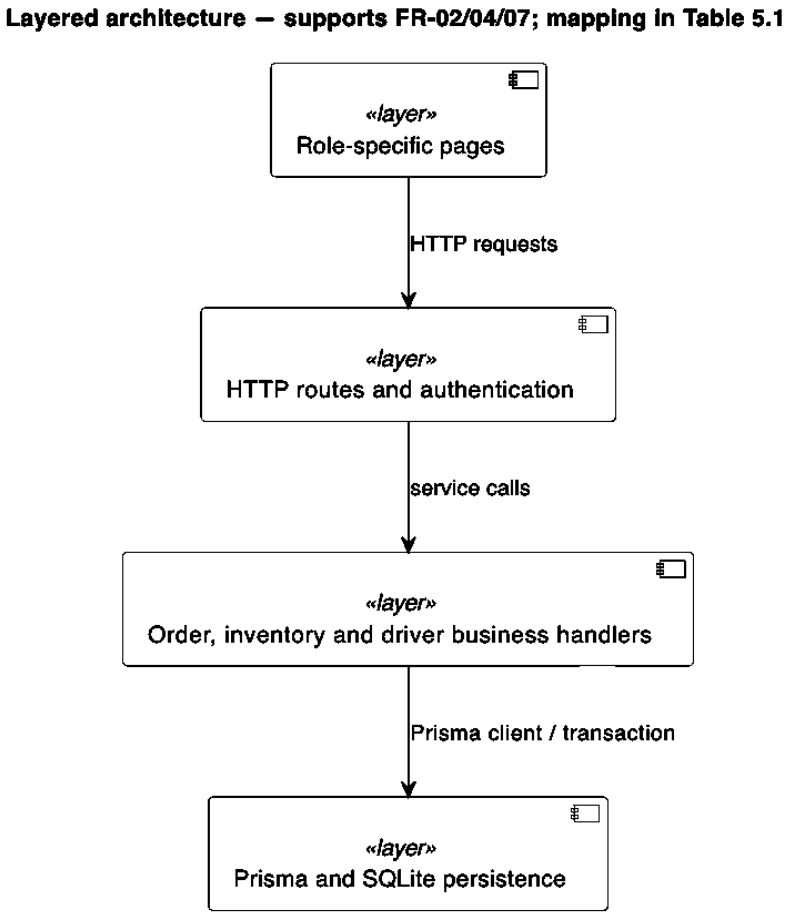


Figure 5. Four layers: role pages call HTTP routes, routes call handlers, handlers use Prisma/SQLite. Source.

5. Implementation

The confirmation transaction reserves stock. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED; CANCELLED from CREATED or CONFIRMED. Cancellations restock unless CREATED. Staff cancellation is restricted to CONFIRMED.

Cancellation permissions:

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CREATED	rejected
Staff	CONFIRMED	cancel, restock
Customer, staff	PICKING onwards	rejected

A CREATED order has not yet been accepted by staff, so staff cannot cancel it.

Key designs and their verification:

Design	Implementation	Verification
Guards: CREATED, stock	order-api.ts:250–253, 276–280	Orders
Confirmation: atomic; rollback on short stock; repeats rejected (409)	order-api.ts:245–293	Orders
driverId: SHIPPED only; kept after delivery	driver-api.ts:44–51; transition	Driver
FR-03 numbering and tracking	create-order.ts:91, shipment-api.ts:62	Logistics chain
FR-04 inventory, ledger, low-stock alerts; staff replenishment adds stock	order-api.ts:281, 398, inventory-api.ts:142	Inventory
FR-05/06/07 handlers	notify.ts:19, reports-api.ts:105, http.ts:58	Notifications, Reports, Auth

6. Verification

CI records 38 unit, 140 integration checks; 2 integrity checks guard the evidence chain, not product behaviour (integrity checks). The pipeline refuses the reset step unless RESET_DEMO_DATA=YES, and the packaged bundle must match the verified source digest (SHA256SUMS.txt). The integration log "tests 140 / pass 140", the unit.log "Tests 38 passed (38)" and the walkthrough record "Browser acceptance passed: 26 checks".

Loopback measurement (11 Nov; 6 endpoints: orders.list, orders.list.filtered, inventory.list, inventory.logs, notifications.list, reports.overview) gave API p95 3.2–13.6 ms and FCP p95 64–96 ms against 2 s and 3 s targets — loopback only, not representative of public networks or high concurrency.

The reviews (Table A rows 2, 6 and 7) also served as static verification of requirements, models and code.

7. Evolution

Three reviews drove the changes: the 17 Oct review found the confirmation defect, fixed by §5; the 10–11 Nov review aligned diagrams and traceability and replaced the NFR-02 availability window with CI-run checks and scoped NFR-01 measurement to the local loopback; the export review of 11 Nov (Table A row 7) added the direct export path (issue #15) and two CI-health regression tests (suite 36 → 38).

8. Validation

The automated walkthrough exercises eleven core operations (subset of the thirteen use cases, Figure 1) — Create an order; Open order details; Confirm order; Start picking; Ship order; Assign driver; Simulate pickup; Simulate delivery; Apply report date filter; Export current report; Export selected dates without applying filter — at most three in-app interface interactions each. "Browser acceptance passed: 26 checks" — 20 operation checks (5 once, 3 twice, 3 thrice), plus 6 cross-cutting checks (two each for input validation, session handling and interface). The walkthrough, its 26 browser checks and the browser.log above are one and the same 11 Nov automated run.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Source	Eviden
Diagrams diverged	Models corrected	10–11 Nov	Correct
Stock effects were recorded under FR-04	Moved stock effects to FR-02; replenishment recording stayed in FR-04	17 Oct	Traceal
Register rows had thin triggers and outcomes	Register rewritten 10–11 Nov; history in git	10–11 Nov	Registe
The filter-then-export path took four interactions	Combined path not adopted; a direct selected-date export was added (three in-app interface interactions); issue closed	11 Nov review	Issue #
No CI-health regression test	Reached 38 tests	11 Nov	Log
NFR-02 daily window	Replaced by CI-run checks (§3) — there is no hosted environment to measure a daily window; each green run verifies the integration points	10–11 Nov	Require
NFR-01 assumed a hosted measurement model	Scoped to loopback measurement; budgets unchanged (§6)	10–11 Nov review	measur

10. Evidence

38 unit, 140 integration, 2 integrity (12 Nov); 26 browser checks (baseline).

Traceability.